

Coding & Solution

Date	02 July 2026
Team ID	xxxxxx
Project Name	Credit Card Approval Prediction Using Machine Learning
Maximum Marks	5 Marks

Coding & Solution

This section presents the implementation details of the Smart Lender project. The application is developed using Python, Flask, and Machine Learning to predict loan eligibility. The codebase follows a modular structure with separate components for data preprocessing, model training, prediction, and the web interface, ensuring maintainability and scalability.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	
Repository Link / URL	https://github.com/24P32A4212/Credit-Card-Approval-Prediction.git
Programming Language(s)	Python, HTML, CSS, JavaScript
Framework(s) Used	Flask, Scikit-learn, Pandas, NumPy, Bootstrap
Key Features Implemented	Data preprocessing, Feature encoding, ML model training, Credit card approval prediction, Web interface using Flask
Pending / Incomplete Features	Model optimization, User authentication, Cloud deployment (if not completed)
Setup / Run Instructions	Install dependencies (pip install -r requirements.txt), run app.py, open http://127.0.0.1:5000 in a browser.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

Credit Card Approval Prediction System

The Credit Card Approval Prediction System is a machine learning–based web application developed using Python, Flask, HTML, CSS, and JavaScript. The system predicts whether a credit card application is likely to be approved or rejected based on applicant details such as income, employment status, credit history, age, and other financial parameters. The application performs input validation, data preprocessing, and prediction using a trained machine learning model. It provides users with instant prediction results through a responsive web interface. The project follows a modular architecture with secure configuration management using environment variables and has been successfully tested and deployed. Future enhancements include user authentication, application history tracking, model retraining with new datasets, explainable AI (XAI) for prediction reasons, and cloud deployment for improved scalability.